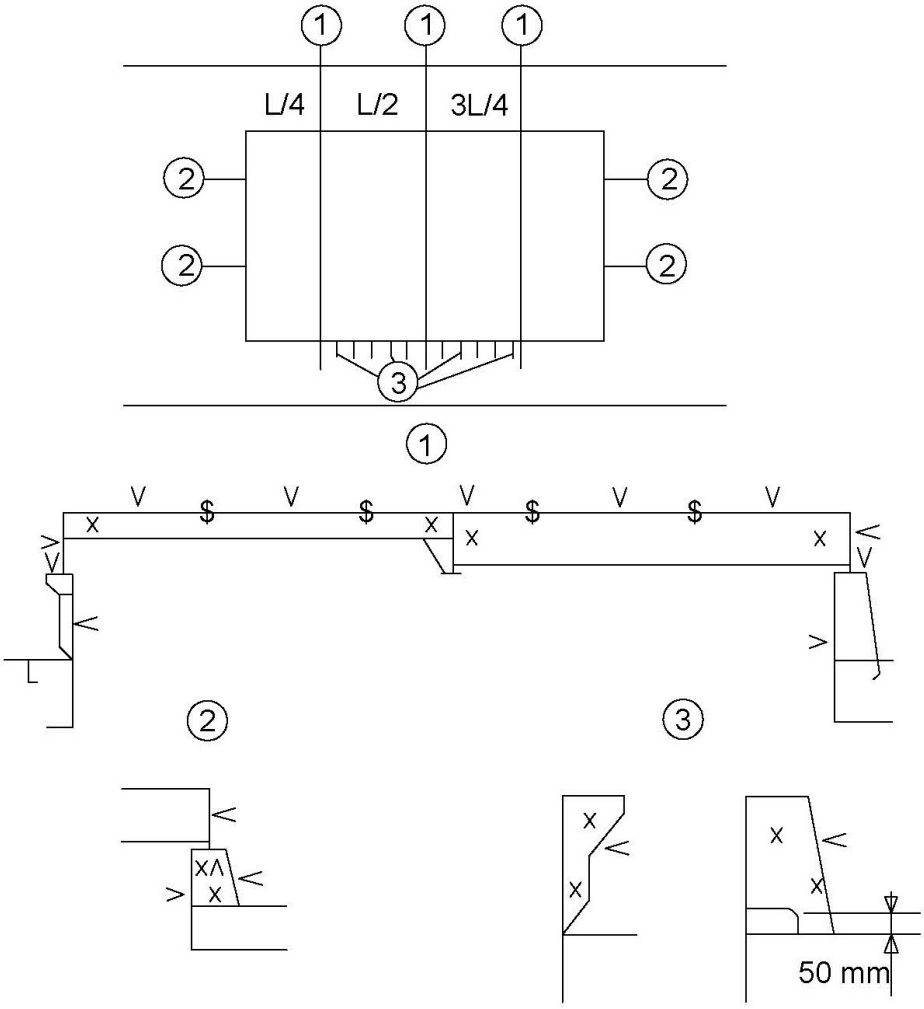
Z10.5
(cont)



Notes:

1. Three sections at L/4, L/2, 3L/4 of hatch cover length, including:

   • one measurement of each hatch cover plate and skirt plate
   • measurements of adjacent beams and stiffeners
   • one measurement of coaming plates and coaming flange, each side

2. Measurements of both ends of hatch cover skirt plate, coaming plate and coaming flange

3. One measurement (two points for web plate and one point for face plate) of one out of three hatch coaming brackets and bars, on both sides and both ends

Figure 5 - Locations of measurements on hatch covers and coamings

Z10.5
(cont)

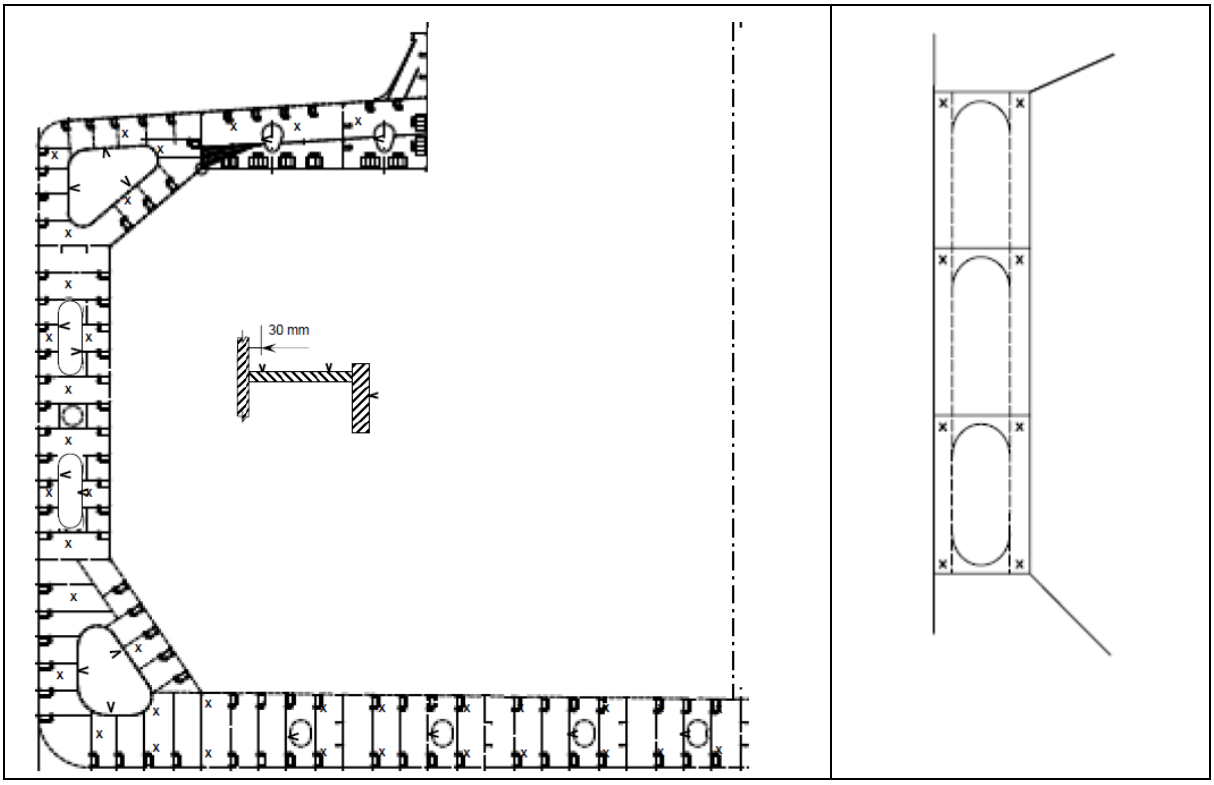
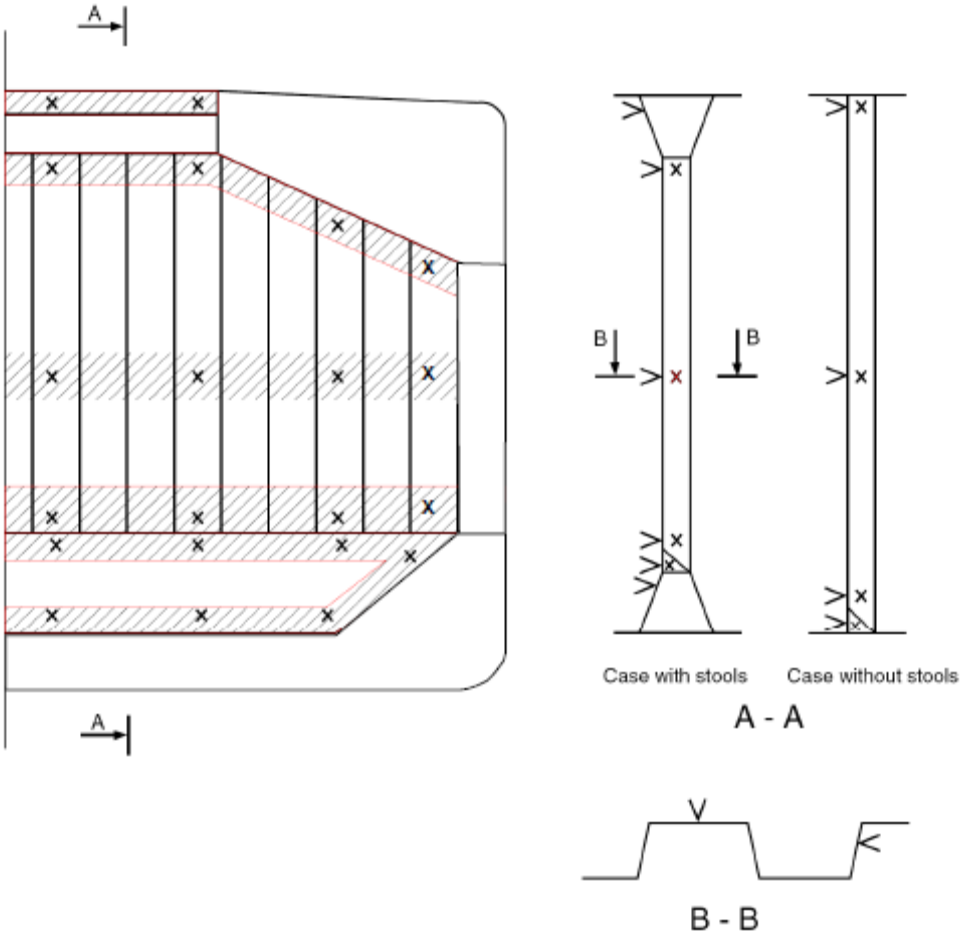


Figure 6 - Locations of measurements on structural members in ballast tanks of double skin bulk carriers (topside or hopper or side tank)

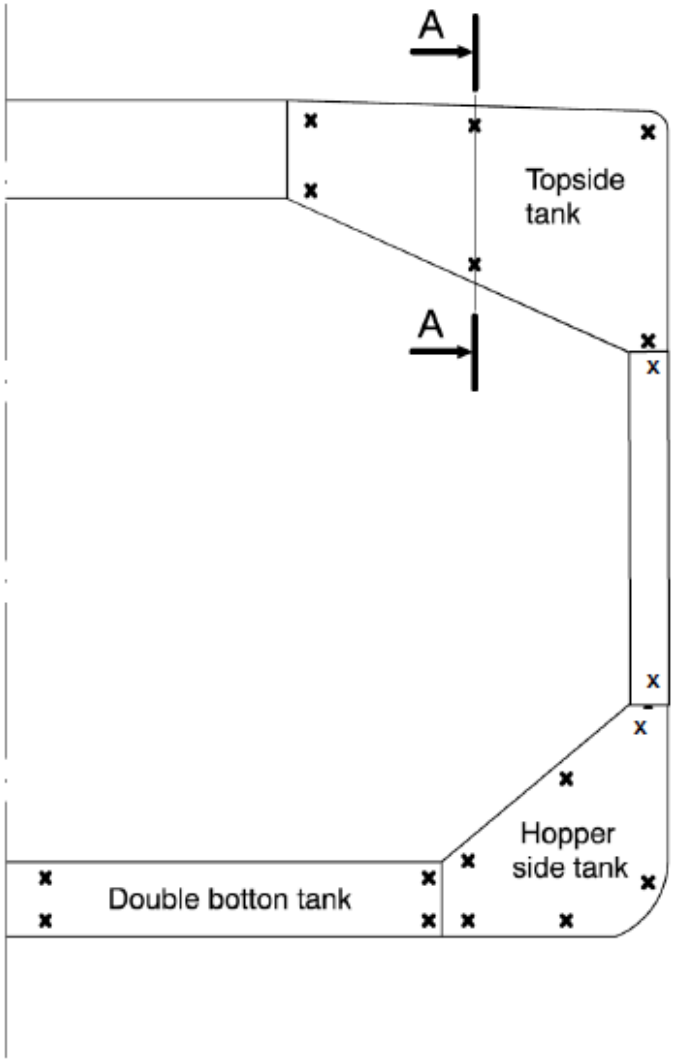
Z10.5
(cont)

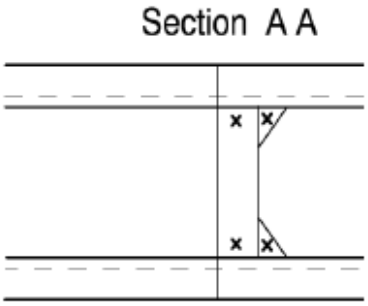


Note: Measurements to be taken in each shaded area as per views A - A and B - B

Figure 7 - Locations of measurements on cargo hold transverse bulkheads (additional measurements to internal structure of upper and lower stools to be added, e.g. two points in the upper and two points in the lower stools to be indicated in section A - A)

Z10.5
(cont)





Note: Measurements to be taken in each vertical section as per view A - A

Figure 8 - Locations of measurements on transverse bulkheads of topside, hopper and double bottom tanks (two additional measurements to internal structure of double bottom tank to be added at midspan)

Z10.5
(cont)

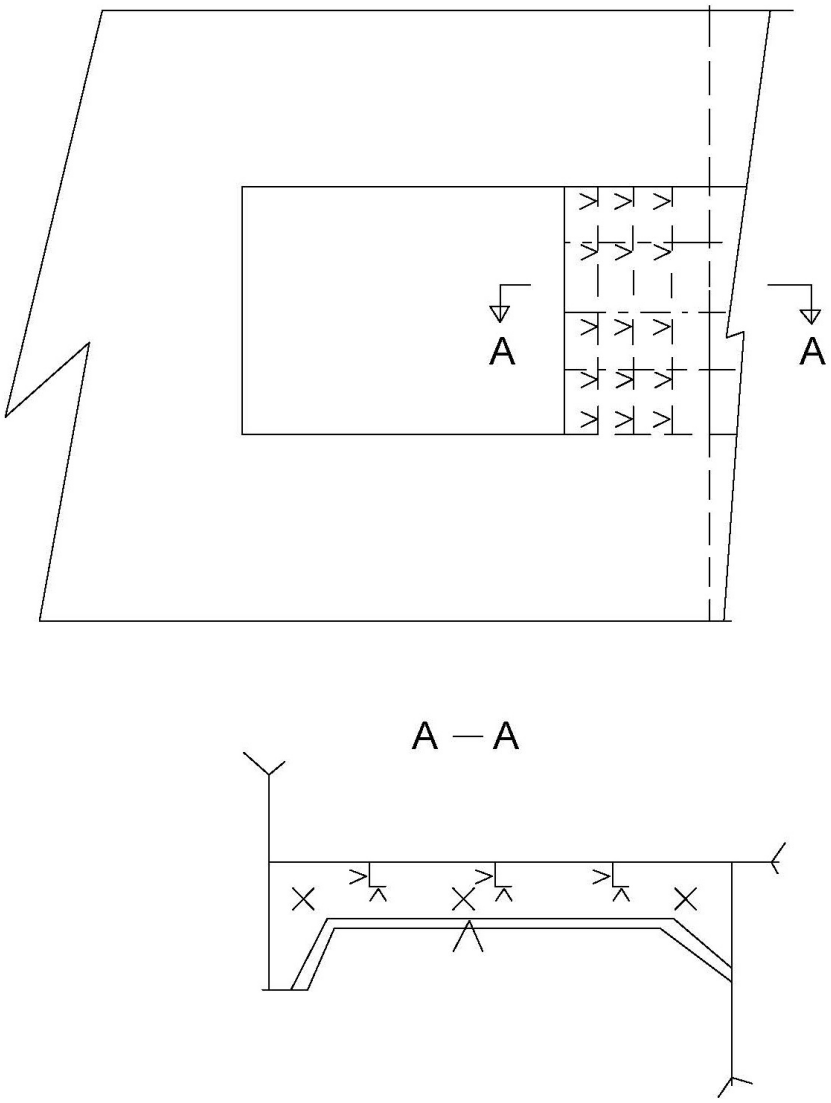


Figure 9 - Locations of measurements on underdeck structure

7.4 Reporting

7.4.1 A thickness measurement report is to be prepared. The report is to give the location of measurement, the thickness measured as well as corresponding original thickness. Furthermore, the report is to give the date when the measurements were carried out, type of measuring equipment, names of personnel and their qualifications and has to be signed by the operator.

The thickness measurement report is to follow the principles as specified in the Recommended Procedures for Thickness Measurement of Double Skin Bulk Carriers, contained in Annex II.

7.4.2 The Surveyor is to review the final thickness measurement report and countersign the cover page.

Z10.5 (cont)

8. ACCEPTANCE CRITERIA

8.1 General

8.1.1 For vessels built under IACS Common Structural Rules, the Acceptance Criteria is according to IACS Common Structural Rules[1] and as specified in 8.2, 8.3 and 8.4.

8.1.2 For vessels not built under IACS Common Structural Rules, the Acceptance Criteria are according to the Rules of the individual Classification Society and/or specific IACS URs depending on ship's age and structural elements concerned, e.g. UR S18 for corrugated transverse watertight bulkheads, UR S19 for the transverse watertight corrugated bulkhead between Cargo Holds Nos. 1 and 2, and UR S21 for all cargo hatch covers and hatch forward and side coamings on exposed decks in position 1 (as defined in ILLC), as applicable.

8.2 Acceptance criteria for pitting corrosion of CSR ships

8.2.1 Side structures

If pitting intensity in an area where coating is required, according to IACS Common Structural Rules[2], is higher than 15% (see Figure 1), thickness measurements are to be performed to check the extent of pitting corrosion. The 15% is based on pitting or grooving on only one side of a plate.

In cases where pitting is exceeding 15%, as defined above, an area of 300 mm or more, at the most pitted part of the plate, is to be cleaned to bare metal and the thickness is to be measured in way of the five deepest pits within the cleaned area. The least thickness measured in way of any of these pits is to be taken as the thickness to be recorded.

The minimum remaining thickness in pits, grooves or other local areas is to be greater than 70% of the as-built thickness, in the side shell, hopper tank and topside tank plating attached to the each side frame, over a width up to 30 mm from each side of it, without being greater than t_{ren}.

8.2.2 Other structures

For plates with pitting intensity less than 20%, see Figure 1, the measured thickness, t_m of any individual measurement is to meet the lesser of the following criteria:

$$t_m \geq 0.7\ (t_{as\text{-}built} - t_{vol\ add})\ \text{mm}$$

$$t_m \geq t_{ren} - 1\ \text{mm}$$

Where:

$t_{as\text{-}built}$ As-built thickness of the member, in mm

$t_{vol\ add}$ Voluntary thickness addition; thickness, in mm, voluntarily added as the Owner's extra margin for corrosion wastage in addition to t_C

t_{ren} Renewal thickness; minimum allowable thickness, in mm, below which renewal of structural members is to be carried out

[1] Ch.13 of IACS CSR for Bulk Carriers, or Ch. 13, Part 1 of IACS CSR BC&OT
[2] Sec. 5, Ch. 3 of IACS CSR for Bulk Carriers, or Sec. 4, Ch. 3, Part 1 of IACS CSR BC&OT

Z10.5 (cont)

t_C Total corrosion addition, in mm, defined in IACS Common Structural Rules[1]

t_m Measured thickness, in mm, on one item, i.e. average thickness on one item using the various measurements taken on this same item during periodical ship's in service surveys.

The average thickness across any cross section in the plating is not to be less than the renewal criteria for general corrosion given in Chapter 13 of CSR.

8.3 Acceptance criteria for edge corrosion of CSR ships

8.3.1 Provided that the overall corroded height of the edge corrosion of the flange, or web in the case of flat bar stiffeners, is less than 25%, see Figure 2, of the stiffener flange breadth or web height, as applicable, the measured thickness, t_m, is to meet the lesser of the following criteria:

$$t_m \geq 0.7\ (t_{as\text{-}built} - t_{vol\ add})\ \text{mm}$$

$$t_m \geq t_{ren} - 1\ \text{mm}$$

8.3.2 The average measured thickness across the breadth or height of the stiffener is not to be less than that defined in IACS Common Structural Rules[2].

8.3.3 Plate edges at openings for manholes, lightening holes etc. may be below the minimum thickness given in IACS Common Structural Rules[2] provided that:

(a) the maximum extent of the reduced plate thickness, below the minimum given in IACS Common Structural Rules[2], from the opening edge is not more than 20% of the smallest dimension of the opening and does not exceed 100mm.

(b) rough or uneven edges may be cropped-back provided that the maximum dimension of the opening is not increased by more than 10% and the remaining thickness of the new edge is not less than t_{ren} - 1 mm.

8.4 Acceptance criteria for grooving corrosion of CSR ships

8.4.1 Where the groove breadth is a maximum of 15% of the web height, but not more than 30mm, see Figure 3, the measured thickness, t_m, in the grooved area is to meet the lesser of the following criteria:

$$t_m \geq 0.75\ (t_{as\text{-}built} - t_{vol\ add})\ \text{mm}$$

$$t_m \geq t_{ren} - 0.5\ \text{mm}$$

but is not to be less than

$$t_m = 6\ \text{mm}$$

8.4.2 Structural members with areas of grooving greater than those in 8.4.1 above are to be assessed based on the criteria for general corrosion as defined in IACS Common Structural Rules[2] using the average measured thickness across the plating/stiffener.

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[1] Sec. 3, Ch. 3 of IACS CSR for Bulk Carriers, or Sec. 3, Ch. 3, Part 1of IACS CSR BC & OT
[2] Ch.13 of IACS CSR for Bulk Carriers, or Ch. 13, Part 1 of IACS CSR BC&OT

Z10.5 (cont)

9 REPORTING AND EVALUATION OF SURVEY

9.1 Evaluation of survey report

9.1.1 The data and information on the structural condition of the vessel collected during the survey is to be evaluated for acceptability and continued structural integrity of the vessel.

9.1.1.1 For CSR bulk carriers, the ship's longitudinal strength is to be evaluated by using the thickness of structural members measured, renewed and reinforced, as appropriate, during the special surveys carried out after the ship reached 15 years of age (or during the special survey no. 3, if this is carried out before the ship reaches 15 years) in accordance with the criteria for longitudinal strength of the ship's hull girder for CSR bulk carriers specified in IACS Common Structural Rules[1].

9.1.1.2 The final result of evaluation of the ship's longitudinal strength required in 9.1.1.1, after renewal or reinforcement work of structural members, if carried out as a result of initial evaluation, is to be reported as a part of the Executive Hull Summary.

9.2 Reporting

9.2.1 Principles for survey reporting are shown in Table VII.

9.2.2 When a survey is split between different survey stations, a report is to be made for each portion of the survey. A list of items examined and / or tested (pressure testing, thickness measurements etc.) and an indication of whether the item has been credited, are to be made available to the next attending Surveyor(s), prior to continuing or completing the survey.

9.2.3 An Executive Hull Summary of the survey and results is to be issued to the Owner as shown in Table VIII and placed on board the vessel for reference at future surveys. The Executive Hull Summary is to be endorsed by the Classification Society's head office or regional managerial office.

[1] Ch.13 of IACS CSR for Bulk Carriers, or Ch. 13, Part 1 of IACS CSR BC&OT

Z10.5
(cont)

TABLE I / Sheet I

MINIMUM REQUIREMENTS FOR CLOSE-UP SURVEY AT SPECIAL HULL SURVEY OF DOUBLE SKIN BULK CARRIERS, EXCLUDING ORE CARRIERS

Special Survey No.1 age ≤ 5	Special Survey No.2 5 < age ≤ 10	Special Survey No.3 10 < age ≤ 15	Special Survey No.4 and subsequent age > 15
One transverse web with associated plating and longitudinals in two representative water ballast tanks of each type. (This is to include the foremost topside and double side water ballast tanks on either side) **(A)**	One tranverse web with associated plating and longitudinals as applicable in each water ballast tank. **(A)** Forward and aft transverse bulkheads including stiffening system in a transverse section including topside, hopper side and double side ballast tanks on one side of the ship (i.e. port or starboard). **(A)** 25% of ordinary transverse frames for transverse framing system or 25% of longitudinals for longitudinal framing system on side shell and inner side plating at forward, middle and aft parts, in the foremost double side tanks. **(B)**	All tranverse webs with associated plating and longitudinals as applicable in each water ballast tank. **(A)** All transverse bulkheads including stiffening system in each water ballast tank. **(A)** 25% of ordinary transverse frames for transverse framing system or 25% of longitudinals for longitudinal framing system on side shell and inner side plating at forward, middle and aft parts, in all double side tanks. **(B)**	All tranverse webs with associated plating and longitudinals as applicable in each water ballast tank. **(A)** All transverse bulkheads including stiffening system in each water ballast tank. **(A)** All ordinary transverse frames for transverse framing system or all of longitudinals for longitudinal framing system on side shell and inner side plating at forward, middle and aft parts, in all double side tanks. **(B)** Areas **(C) – (E)** as for age interval 10 to 15 years.
Two selected cargo hold transverse bulkheads, including internal structure of upper and lower stools, where fitted. **(C)**	One transverse bulkhead in each cargo hold, including internal structure of upper and lower stools, where fitted. **(C)**	All cargo hold transverse bulkheads, including internal structure of upper and lower stools, where fitted. **(C)**	
All cargo hold hatch covers and coamings (platings and stiffeners). **(D)**	All cargo hold hatch covers and coamings (platings and stiffeners). **(D)**	All cargo hold hatch covers and coamings (platings and stiffeners). **(D)**	

Z10.5
(cont)

	All deck plating and under deck structure inside line of hatch openings between all cargo hold hatches. **(E)**	All deck plating and under deck structure inside line of hatch openings between all cargo hold hatches. **(E)**	

(A), (B), (C), (D) and **(E)** are areas to be subjected to close-up surveys and thickness measurements (see Figure 10 and 11).

(A): Transverse web frame or watertight transverse bulkhead in topside, hopper side and double side ballast tanks. In fore and aft peak tanks transverse web frame means a complete transverse web frame ring including adjacent structural members

(B): Ordinary transverse frame in double side tanks

(C): Cargo hold transverse bulkheads plating, siffeners and girders

(D): Cargo hold hatch covers and coamings. Subject to cargo hold hatch covers of approved design which structurally have no access to the internals, close-up survey/thickness measurement shall be done of accessible parts of hatch covers structures.

(E): Deck plating and under deck structure inside line of hatch openings between cargo hold hatches

Note: Close-up survey of transverse bulkheads to be carried out at four levels:

Level (a): Immediately above the inner bottom and immediately above the line of gussets (if fitted) and shedders for ships without lower stool.

Level (b): Immediately above and below the lower stool shelf plate (for those ships fitted with lower stools), and immediately above the line of the shedder plates.

Level (c): Above mid-height of the bulkhead.

Level (d): Immediately below the upper deck plating and immediately adjacent to the upper wing tank, and immediately below the upper stool shelf plate for those ships fitted with upper stools, or immediately below the topside tanks.

Z10.5
(cont)

TABLE I / Sheet 2

MINIMUM REQUIREMENTS FOR CLOSE-UP SURVEY AT SPECIAL HULL SURVEY OF ORE CARRIERS

Special Survey No.1 age ≤ 5	Special Survey No.2 5 < age ≤ 10	Special Survey No.3 10 < age ≤ 15	Special Survey No.4 and subsequent age > 15
One web frame ring complete including adjacent structural members in a ballast wing tank. **(A)** One transverse bulkhead lower part - including girder system and adjacent structural members - in a ballast tank. **(A)**	All web frame rings complete including adjacent structural members in a ballast wing tank. **(A)** One deck transverse including adjacent deck structural members in each remaining ballast tank. **(A)** Forward and aft transverse bulkheads complete - including girder system and adjacent structural members - in a ballast wing tank. **(A)** One transverse bulkhead lower part - including girder system and adjacent structural members - in each remaining ballast tank. **(A)**	All web frame rings complete including adjacent structural members in each ballast tank. **(A)** All transverse bulkheads complete - including girder system and adjacent structural members - in each ballast tank. **(A)** One web frame ring complete including adjacent structural members in each wing void space. **(A)** Additional web frame rings in void spaces as deemed necessary by the Classification Society. **(A)**	As for Special Survey for age from 10 to 15 years.
Two selected cargo hold transverse bulkheads, including internal structure of upper and lower stools, where fitted. **(C)**	One transverse bulkhead in each cargo hold, including internal structure of upper and lower stools, where fitted. **(C)**	All cargo hold transverse bulkheads, including internal structure of upper and lower stools, where fitted. **(C)**	Areas **(C) - (E)** as for age interval 10 to 15 years.
All cargo hold hatch covers and coamings (plating and stiffeners). **(D)**	All cargo hold hatch covers and coamings (plating and stiffeners). **(D)**	All cargo hold hatch covers and coamings (plating and stiffeners). **(D)**	
	All deck plating and under deck structure inside line of hatch openings between all cargo hold hatches. **(E)**	All deck plating and under deck structure inside line of hatch openings between all cargo hold hatches. **(E)**	

Z10.5
(cont)

(A), (C), (D) and **(E)** are areas to be subjected to close-up surveys and thickness measurements (see Figure 10 and Figure 11).

(A): Transverse web frame or watertight transverse bulkhead in ballast wing tanks and void spaces. In fore and aft peak tanks transverse web frame means a complete transverse web frame ring including adjacent structural members

(C): Cargo hold transverse bulkheads plating, stiffeners and girders

(D): Cargo hold hatch covers and coamings. Subject to cargo hold hatch covers of approved design which structurally have no access to the internals, close-up survey/thickness measurement shall be done of accessible parts of hatch covers structures.

(E): Deck plating and under deck structure inside line of hatch openings between cargo hold hatches

Note: Close-up Survey of transverse bulkheads to be carried out at four levels:

Level (a): Immediately above the inner bottom and immediately above the line of gussets (if fitted) and shedders for ships without lower stool.

Level (b): Immediately above and below the lower stool shelf plate (for those ships fitted with lower stools), and immediately above the line of the shedder plates.

Level (c): About mid-height of the bulkhead.

Level (d): Immediately below the upper deck plating and immediately adjacent to the upper wing tank, and immediately below the upper stool shelf plate for those ships fitted with upper stools, or immediately below the topside tanks.

Z10.5
(cont)

TABLE II

MINIMUM REQUIREMENTS FOR THICKNESS MEASUREMENTS AT SPECIAL HULL SURVEY OF DOUBLE SKIN BULK CARRIERS

Special Survey No.1 age ≤ 5	Special Survey No.2 5 < age ≤ 10	Special Survey No.3 10 < age ≤ 15	Special Survey No.4 and subsequent age > 15
Suspect areas	Suspect areas	Suspect areas	Suspect areas
	Within the cargo length: - Two transverse sections of deck plating outside line of cargo hatch openings	Within the cargo length: - each deck plate outside line of cargo hatch openings - two transverse sections, one in the amidship area, outside line of cargo hatch openings - all wind and water strakes	Within the cargo length: - each deck plate outside line of cargo hatch openings - three transverse sections, one in the amidship area, outside line of cargo hatch openings - each bottom plate
	Wind and water strakes in way of the two transverse sections considered above Selected wind and water strakes outside the cargo length area	Selected wind and water strakes outside the cargo length area	All wind and water strakes, full length
	Measurements, for general assessment and recording of corrosion pattern, of those structural members subject to close-up survey according to Table I/Sheet 1 or Table I/Sheet 2, as applicable.	Measurements, for general assessment and recording of corrosion pattern, of those structural members subject to close-up survey according to Table I/Sheet 1 or Table I/Sheet 2, as applicable.	Measurements, for general assessment and recording of corrosion pattern, of those structural members subject to close-up survey according to Table I/Sheet 1 or Table I/Sheet 2, as applicable.

Z10.5
(cont)

TABLE III

Ship Name: .....

OWNERS INSPECTION REPORT - Structural Condition
For Tank/Hold No: .....

Grade of Steel:	Deck	:	Side	: .....
	Bottom	:	Long. Bhd	: .....

Elements Other	Cracks	Buckles	Corrosion	Coating cond.	Pitting	Mod. /Rep.

Deck:

Bottom:

Side:

Long. Bulkheads:

Transv. Bulkheads:

Repairs carried out due to:

Thickness measurements carried out, dates:
Results in General:

Overdue Surveys:

Outstanding Conditions of class:

Comments:

*Repairs are to be surveyed by the classification society

Date of Inspection: _____
Inspected by: _____
Signature: _____

Z10.5
(cont)

TABLE IV/ Sheet 1

REQUIREMENTS FOR EXTENT OF THICKNESS MEASUREMENTS AT THOSE AREAS OF SUBSTANTIAL CORROSION OF DOUBLE SKIN BULK CARRIERS WITHIN THE CARGO LENGTH AREA

BOTTOM, INNER BOTTOM AND HOPPER STRUCTURE		
Structural member	Extent of measurement	Pattern of measurement
Bottom, inner bottom and hopper structure plating	Minimum of three bays across double bottom tank, including aft bay Measurements around and under all suction bell mouths	Five-point pattern for each panel between longitudinals and floors
Bottom, inner bottom and hopper structure longitudinals	Minimum of three longitudinals in each bay where bottom plating measured	Three meaurements in line across the flange and three measurements on the vertical web
Bottom girders, including the watertight ones	At fore and aft watertight floors and in centre of tanks	Vertical line of single measurements on girder plating with one measurement between each panel stiffener, or a minimum of three measurements
Bottom floors, including the watertight ones	Three floors in the bays where bottom plating measured, with measurements at both ends and middle	Five-point pattern over two square metre area
Hopper structure web frame ring	Three floors in bays where bottom plating measured	Five-point pattern over one square metre of plating Single measurements on flange
Hopper structure transverse watertight bulkhead or swash bulkhead	- lower 1/3 of bulkhead	- five-point pattern over one square metre of plating
	- upper 2/3 of bulkhead	- five-point pattern over two square metre of plating
	- stiffeners (minimum of three)	- For web, five-point pattern over span (two measurements across web at each end and one at centre of span). For flange, single measurements at each end and centre of span
Plate stiffening	Where applicable	Single measurements